Next Generation Data Centers and Energy Management Systems

Lecture 11A: Modeling and Simulation of an Energy Management System

Prof. Thomas M. Schutzius

Department of Mechanical Engineering

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Berkeley
UNIVERSITY OF CALIFORNIA

Learning Plan

- Introduction to Data Center Systems and Thermodynamics
- Psychrometrics and Heat Transfer
- Performance Metrics and Standards
- Applied Computational Fluid Dynamics (CFD)
- Liquid Cooling & Advanced Heat Transfer
- Chip Cooling: Heat Conduction and Thermal Interface Materials
- Power Delivery and Infrastructure (Energy sources: Geothermal, Solar, Nuclear, etc.)
- Overview of Simulation Field; the Concept of a Digital Twin
- Machine Learning and Optimization Tools
- **Modeling and Simulation of an Energy Management System**
- **Control Systems and Optimization**
- Spacecraft Thermal Control
- Final Project Work, Peer Feedback, and Mentor Check-in
- Symposium

Learning Objectives

- Discuss expectations for the final project
- Apply classical control, model predictive control, and reinforcement learning to a cooling distribution unit.

Typical layout of a data center facility

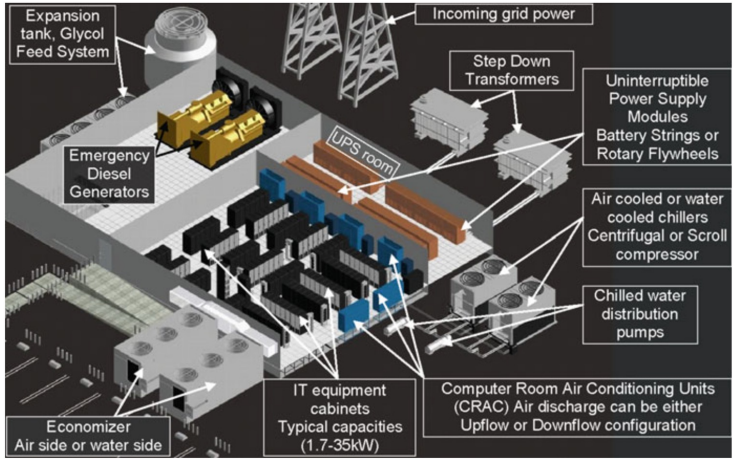


Figure 1: (Figure 1.1 in Ref. [2]).

Seminar 9

- Bryson Borzini
- Title: Building Next-Generation Data Centers: A Mechanical Contractor’s Perspective
- Date: November 17, 2026
- Time: 15:30 (Berkeley Time)
- Location: GSPP 150

Modeling and Predictive Control of a Rack CDU

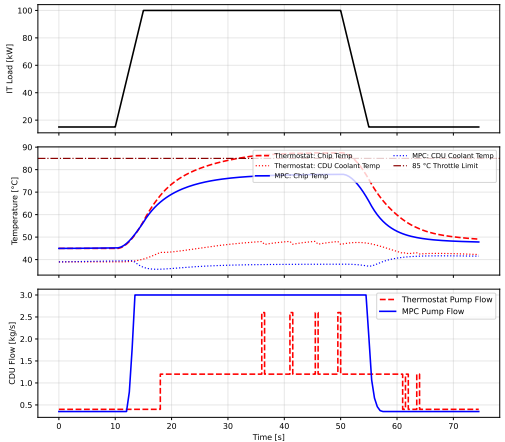


Figure 2: CDU thermal management during an idle to full-throttle process.

Reinforcement Learning Control of a Rack CDU

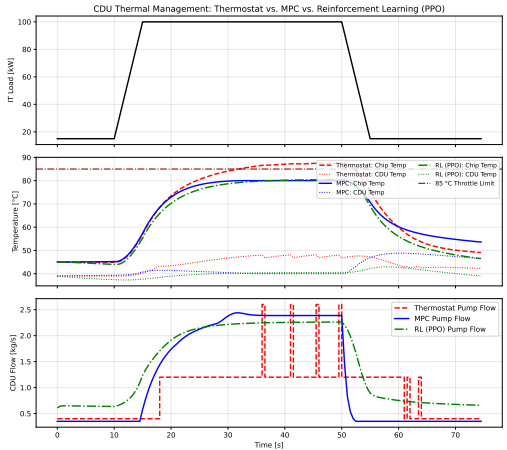


Figure 3: CDU thermal management during an idle to full-throttle process.

References I

[1] Jason Brownlee.
Deep learning for time series forecasting: predict the future with MLPs, CNNs and LSTMs in Python.
Machine Learning Mastery, 2018.

[2] Yogendra Joshi and Pramod Kumar, editors.
Energy Efficient Thermal Management of Data Centers.
Springer US, 2012.